AIML MINOR PROJECT 2

Predict New York City Airbnb Prices Using Linear Regression

1. Objective

The objective of this project is to build a Machine Learning model using Linear Regression to predict Airbnb listing prices in New York City based on features such as location, room type, minimum nights, number of reviews, and availability.

2. Dataset Description

The project uses the New York City Airbnb Open Data dataset from Kaggle. It contains approximately 49,000 listings with numerical and categorical features describing Airbnb properties.

3. Data Preprocessing

Exploratory Data Analysis (EDA) was performed to understand the dataset's structure, data types, and missing values. Irrelevant columns such as listing IDs, host details, listing names, neighbourhood, and last review dates were removed. Missing values in `reviews_per_month` were filled with zero for listings with no reviews. Outliers in `price` and `minimum_nights` were treated using the Interquartile Range (IQR) method with clipping. Categorical features, including `neighbourhood_group` and `room_type`, were converted into numerical values using One-Hot Encoding with `drop_first=True`.

4. Model Development

The dataset was divided into training and testing sets using an 80:20 ratio, with `random_state=42` to ensure reproducibility. A Linear Regression model was trained on the training data and used to predict prices for the testing data.

5. Model Evaluation

The model was evaluated using the following metrics:

- **R-squared (R^2):** [Insert your R^2 score]
- **Mean Absolute Error (MAE):** [Insert your MAE value in USD]
- **Mean Squared Error (MSE):** [Insert your MSE value]
- **Root Mean Squared Error (RMSE):** [Insert your RMSE value in USD]

Actual-versus-predicted price plots and residual plots were used to analyze prediction accuracy and errors. The MAE comparison showed that the model performed better for [cheaper/more expensive] listings, based on the lower average absolute prediction error.

6. Conclusion

This project demonstrates the application of Linear Regression for predicting Airbnb prices using numerical and encoded categorical features. Data cleaning, outlier treatment, and feature encoding were important preprocessing steps. The evaluation metrics and visualizations helped assess the model's performance and identify its limitations. The results can help understand the factors associated with Airbnb listing prices, although the model's predictions may not capture every factor influencing actual prices.